3.3.1 Number of research papers published per teacher in the journals notified on UGC Care list during the last five years

2019

List of paper publications

S.NO	Title of the paper	Name of the author/s	Name of the journal	Page No
1	A New and precise Bioanalytical method development and validation for the estimation of efavirenz in human plasma by RP-HPLC	**KS.Sumanth,** A.Srinivasarao, D.Gowrishankar	International journal of pharmaceutical, chemical and biological sciences.	4-6

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Research Article

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A NEW AND PRECISE BIO-ANALYTICAL METHOD DEVELOPMENT AND VALIDATION FOR THE ESTIMATION OF EFAVIRENZ IN HUMAN PLASMA BY RP-HPLC

KS. Sumanth[1]*, A. Srinivasa Rao[2] and D. Gowri Shankar[3]

[1]Department of Pharmaceutical Analysis, Sri Vasavi Institute of Pharmaceutical Sciences, Andhra Pradesh, India.

[2]Department of Pharmaceutical Analysis, Shri Vishnu College of Pharmacy,Andhra Pradesh, India.

[3]University College of Pharmaceutical Sciences, Andhra University, Visakhapatnam, Andhra Pradesh, India.

ABSTRACT

A new, simple, rapid and accurate RP-HPLC method was developed for the estimation of Efavirenz (EFA) in human plasma and validated. The method was developed on SHISEIDO C18 column (250 x 4.6 mm i.d, 5µ) using Acetonitrile : 20 mM phosphate buffer (pH 3.0). The wavelength used for detection of EFA is at 247 nm. The flow rate used for the elution is 1 ml/min. sample preparation for the extraction of drug was done using solid phase extraction method by Phenomenex STRATA® C18 sorbent, a 24 station SPE vacuum extraction assembly is used for the entire method. M̅ethyl prednisolone was used as internal standard, the retention time of Efavirenz and internal standard were found to be 5.7 and 3.7 min respectively. Linearity was established in the range of 0.43 8.60 µg/ml and the coefficient of regression (R2) value was found to be 0.995, having 0.037 as slope. The method was precise with %RSD < 2 for both intraday and interday precision. The accuracy of the method was performed over three levels of concentration and the recovery was in the range of 98-102%. The method can be successfully applied for quantifying the drug in human plasma for bioavailability and bioequivalence studies.

INTRODUCTION

Efavirenz[1,2] (6-chloro- 4 - (2 – Cyclopropyl – 1 - ethynyl) – 4 – trifluoro methyl (4S) - 1, 4 – – dihydro - 2H benzo [D] [1,3] oxazin - 2-one) isa non-nucleoside reverse transcriptase inhibitor (NNRTI) and is used as part of highly active antiretroviral therapy (HAART) for the treatment of a human immunodeficiency virus (HIV) type 1. For HIV infection that has not previously been treated, Efavirenz and Lamivudine in combination with Zidovudine or Tenofovir is the preferred NNRTI-based regimen. Efavirenz is also used in combination with other antiretroviral agents as part of an expanded post exposure prophylaxis regimen to prevent HIV transmission for those exposed to materials associated with a high risk for HIV transmission[3,4] (Fig. 1)

The present study is to develop a RP-HPLC method for the determination of EFA in human plasma. The Literature survey reports different analytical methods for EFA based on UV[5-7], HPTLC[8], HPLC[9-15], stability indicating HPLC[16-19], Bio-analytical[20,21], LC-MS[22,23] were reported. However there were few methods reported for the determination of EFA in huan plasma, the present aim is to develop a more precise, accurate, simple and RP-HPLC method for the estimation of EFA in human plasma and validate according to USFDA guidelines. The molar absorptivity of EFA was found maximum at 247 nm. The validated method will be used for the

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